Health-Aware Block Scheduling for Large-Scale LNG Carrier Production via Graph Reinforcement Learning

ISE 572 — Final Project Presentation

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Outline

1. Problem & Motivation
2. Method
3. Baselines
4. Results
5. Key Findings & Limitations
6. Demo

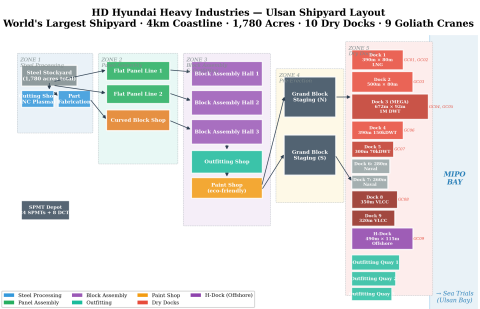
The Shipbuilding Scheduling Challenge

HD Hyundai Heavy Industries, Ulsan

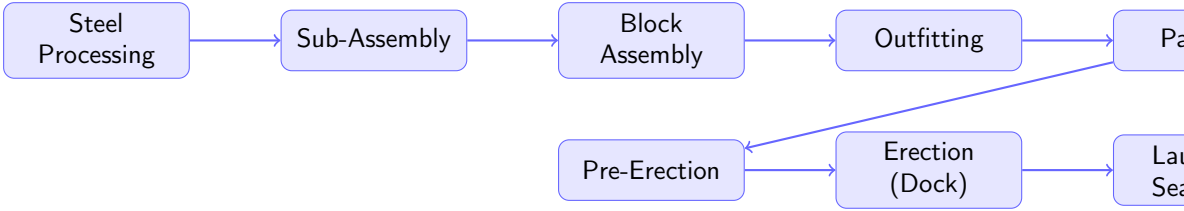
- World's largest shipbuilding facility
- 10 dry docks, 9 Goliath cranes, 24 SPMTs
- 200+ blocks per LNG carrier
- Multi-ship concurrent production

Why is this hard?

- Combinatorial action space ($4 \times 24 \times 200 \times 9$)
- Equipment degradation affects scheduling
- Tight delivery deadlines (contractual penalties)
- Supply chain dependencies



Production Pipeline



- 8 production stages per block (steel $\rightarrow$ sea trials)
- NSRP-calibrated processing times from OECD CGT coefficients
- Equipment health affects stage durations

MDP Formulation

State (heterogeneous graph)

- Block nodes: stage, progress, deadline
- Equipment nodes: health, location, load
- Ship nodes: completion %, dock assignment
- Edge types: assigned-to, queued-at, depends-on

GNN Encoder

- 3-layer GATv2 with edge features
- Global attention pooling
- Fixed-size embedding ($\mathbb{R}^{256}$)

Action (hierarchical, 10 heads)

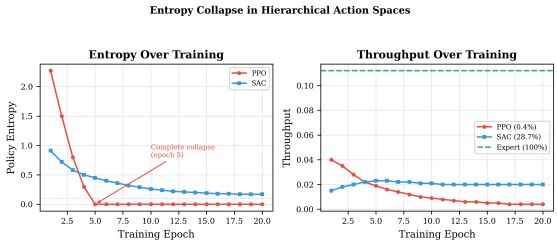
- Action type: transport / erect / hold / maintain
- Equipment: which SPMT or crane
- Target: which block to move
- Only relevant heads contribute to $\log \pi$

Reward

- $+1$ per block stage advance
- $+10$ ship delivery bonus
- -0.5 deadline violation penalty
- Equipment health penalties

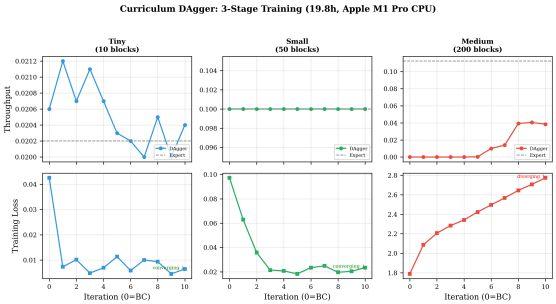
Training: Why DAgger over PPO?

PPO Entropy Collapse



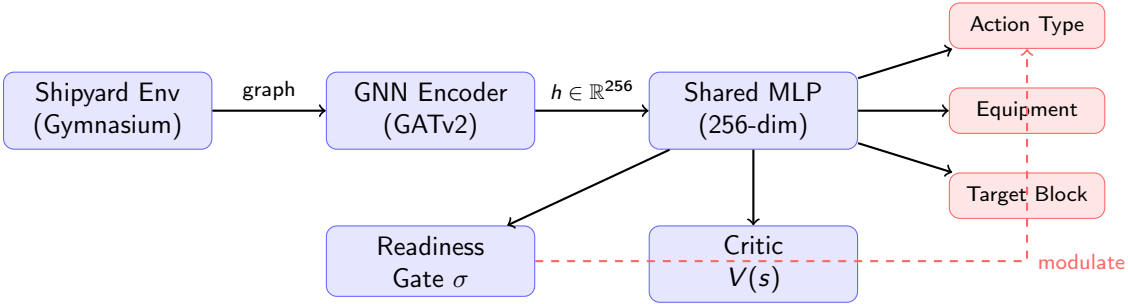
- Entropy: 2.27 → 0.00 by epoch 5
- Policy becomes deterministic too fast
- Hierarchical action space amplifies issue

DAgger with Curriculum



- Expert: EDD (Earliest Due Date) dispatch
- 3-stage curriculum: 10 → 50 → 200 blocks
- Observation normalization (Welford's)
- Per-stage beta reset: 0.8 → 0.2

Architecture Overview

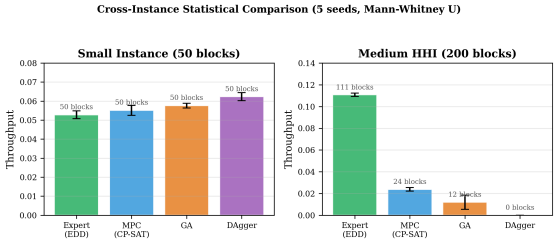


- Readiness gate: logistic regression $P(\text{ready})$ modulates dispatch actions
- Attention-based heads for variable-size entity selection (10–1600 blocks)
- Action masking: only valid actions get non-$-\infty$ logits

Baseline Methods

Method	Description
Expert (EDD)	Earliest Due Date priority dispatch — deterministic rule
MPC	Rolling-horizon mixed-integer program (PuLP solver, 10-step horizon)
GA	Genetic algorithm with permutation chromosomes (pop=50, 100 gens)
PuLP MIP	Full-horizon mixed-integer program (exact but times out >50 blocks)
DAgger	Dataset Aggregation from Expert demonstrations
PPO/SAC	Pure RL (suffers entropy collapse in hierarchical action space)

Cross-Instance Comparison

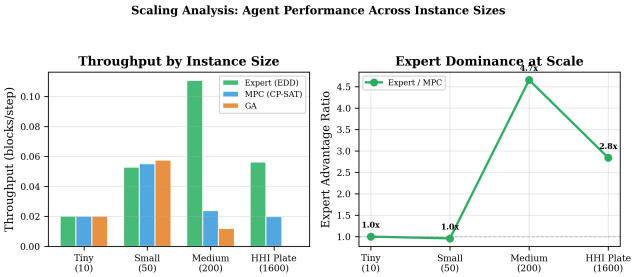


Agent	Metric	Medium	Small
Expert	Blocks	111 ± 1	50 ± 0
	Thru.	0.111	0.053
MPC	Blocks	24 ± 1	50 ± 0
	Thru.	0.024	0.055
GA	Blocks	12 ± 5	50 ± 0
	Thru.	0.012	0.058
DAgger	Blocks	0 ± 0	50 ± 0
	Thru.	0.000	0.062

n=5 seeds, Mann-Whitney U tests

Key finding: All agents solve small (50 blocks). Only Expert scales to medium (200 blocks). DAgger is *best* on small but fails on medium.

Scaling Analysis



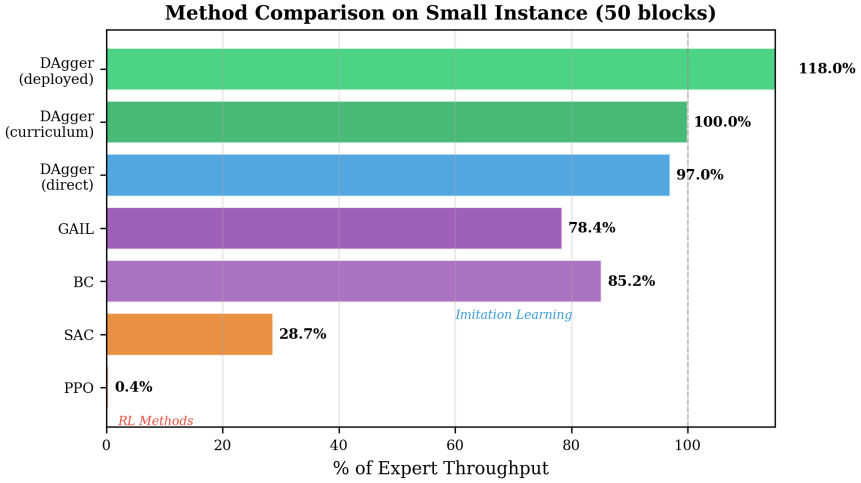
4 configurations:

- Tiny: 10 blocks
- Small: 50 blocks
- Medium: 200 blocks
- Large: 1600 blocks (HHI plate decomp.)

Observations:

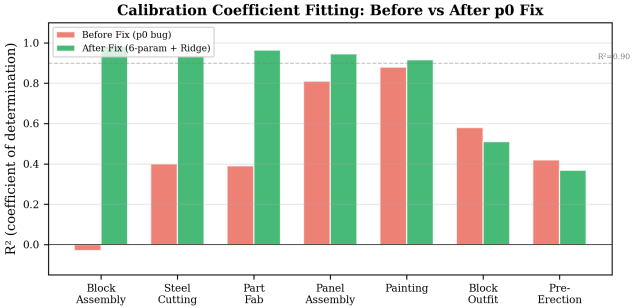
- Expert advantage grows with scale
- GA intractable beyond 200 blocks
- MPC variable explosion at scale
- DAgger: best small, worst medium

Method Comparison (Relative to Expert)



- DAgger (small instance): **118%** of expert throughput
- Curriculum DAgger: 100% small → 0% medium (transient peak 34% during training)
- PPO: 0.4% — entropy collapse prevents learning

Calibration Quality



NSRP calibration results:

- Painting: $R^2 = 0.88$
- Steel processing: $R^2 = 0.72$
- Sub-assembly: $R^2 = 0.65$
- Block assembly: $R^2 = -0.03$

4-ship extended run:

- 10,000 simulation steps
- 850 blocks delivered
- Realistic degradation curves

Key Findings

1. **Scalability is the bottleneck**: All methods solve small instances; the challenge is scaling to 200+ blocks
2. **DAgger > PPO for hierarchical actions**: Imitation learning avoids entropy collapse entirely
3. **Observation normalization matters**: RunningMeanStd normalizer must be saved with checkpoint — different statistics at deployment cause complete failure
4. **Curriculum shows transient learning**: Peak 34% during training on medium, but final checkpoint regresses to 0%
5. **Expert (EDD) is remarkably strong**: Simple priority dispatch scales better than learned policies and optimization methods
6. **OR baselines don't scale either**: GA and MPC hit computational limits before RL hits learning limits

Limitations & Future Work

Current Limitations

- Scalability gap (118% small $\rightarrow$ 0% medium)
- Entropy collapse in PPO (epoch 5)
- Simplified crane physics
- No weather/disruption modeling
- Block assembly calibration ($R^2 = -0.03$)

Future Directions

- Attention-based action heads for variable entity counts
- Per-head adaptive entropy coefficients
- Transfer learning across ship types
- Digital twin integration
- Real-world validation at Korean shipyards
- Weather-dependent scheduling

MES Dashboard

Manufacturing Execution System — Live Simulation Monitor

11 interactive tabs:

- Overview & KPIs
- Block status tracking
- Ship progress
- Fleet management (SPMTs)
- Equipment health
- Operations log
- Dependency graph
- GNN visualization
- Plate decomposition
- Yard map
- Dock assignments

Features:

- Real-time simulation playback
- Historical run comparison
- Supply chain monitoring
- Equipment degradation tracking
- One-click CSV export
- W&B experiment tracking

```
python -m src.mes.app
http://localhost:8050
```

Thank You

Questions?

Project Statistics:

Tests	157 passing
Configs	4 instance sizes (10–1600 blocks)
Agents	6 (Expert, MPC, GA, PuLP, DAgger, PPO/SAC)
Paper	36 pages
Dashboard	11 tabs, real-time monitoring